EXPERIMENT – 2

NAME: VEDHASREE S

REGISTER NUMBER: 240701580

Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1: INSTRUCTION PAGE

Chunking Analysis of the Instruction Page

Chunking is a cognitive technique that organizes information into smaller, meaningful units to reduce cognitive load. The instruction screen of the Visual Memory Recall Task applies chunking effectively in the following ways:

1. Clear and Sequential Instructions

- Instructions are presented in numbered steps, guiding the user through the task in a clear order.
- Each step focuses on one specific action, preventing confusion and overload.

2. Logical Flow of Information

The instructions are grouped into logical phases:

- Observation Phase (Steps 1–2): Explains that icons will be shown and observed for 10 seconds.
- Recall Phase (Steps 3–5): Describes hiding the icons and selecting remembered items.
- Evaluation Phase (Steps 6–7): Explains submission and score display.

This grouping helps users mentally prepare for each stage of the task.

3. Visual Hierarchy and Layout

- A clear title ("Visual Memory Recall Task") immediately communicates the purpose of the screen.
- The instruction text is placed inside a separate container, visually chunking it from the background.
- Adequate spacing between lines improves readability.
- A prominent "Start" button signals the next action clearly.

4. Time Awareness

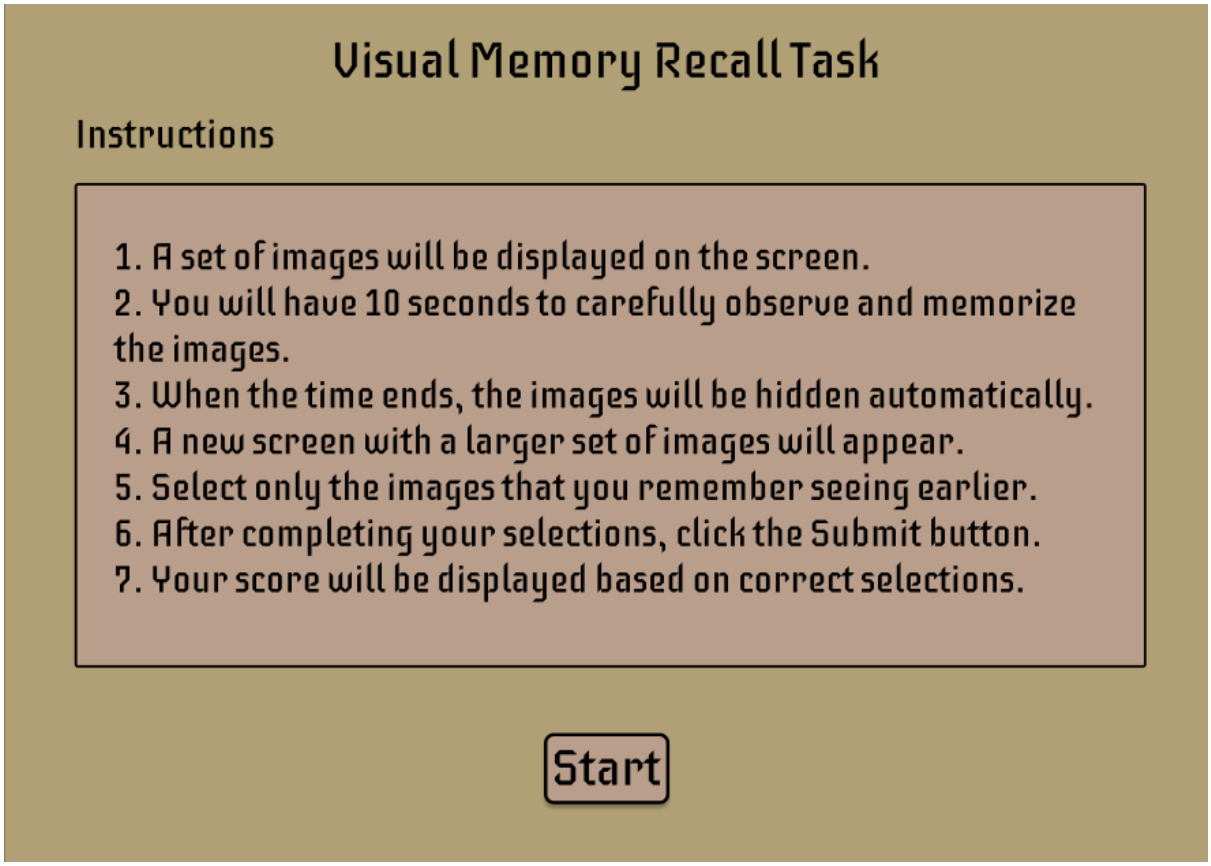
- The instruction explicitly states that users have 10 seconds to observe the icons.

- This sets expectations and encourages focused attention.

5. Simplicity and Clarity

- Instructions use short, direct sentences.

Simple language ensures that users of all skill levels can understand the task easily.



FRAME 2: OBSERVATION (CHUNKING) PHASE

Analysis of the Observation Phase Screen

This screen represents the visual encoding stage, where users observe and memorize icons arranged in a structured layout.

1. Purpose of the Screen

- This phase allows users to encode visual information into short-term memory.
- The layout encourages the use of chunking strategies such as grouping similar objects.

2. Key UI Elements

Grid of Icons (3 × 6 Layout)

- Icons such as laptop, phone, headphones, watch, printer, camera, book, car, apple, and tree are displayed.

- The 3×6 grid prevents overcrowding and supports visual scanning.
- Users can chunk icons by:
    - Category (electronics, objects, nature)
    - Shape or familiarity
    - Mental grouping of rows or columns

Time Constraint (10 Seconds)

- A fixed observation time encourages quick pattern recognition.
- Prevents over-analysis and promotes natural memory strategies.

3. Chunking Benefits in This Phase

- Structured grid layout helps users remember icons row-wise or column-wise.
- Familiar real-world objects improve recall accuracy.
- Reduces cognitive load by presenting information in a clean, organized manner.

FRAME 3: RECALL / SELECTION PHASE

Analysis of the Recall Phase Screen

This screen evaluates the user’s ability to retrieve information from memory after the observation phase.

1. Purpose of the Screen

- Tests memory retrieval by asking users to identify icons seen earlier.
- Measures how effectively chunking helped retain visual information.

2. Key UI Components

Instruction Text

- “Select the items you remember” clearly communicates the task goal.

Expanded Grid of Icons

- Contains both:
    - Previously shown icons (correct answers)
    - New icons (distractors)
- Icons are mixed randomly to prevent position-based guessing.

Selection Indicators

- Selected icons are visually marked, giving users confidence in their choices.

Submit Button

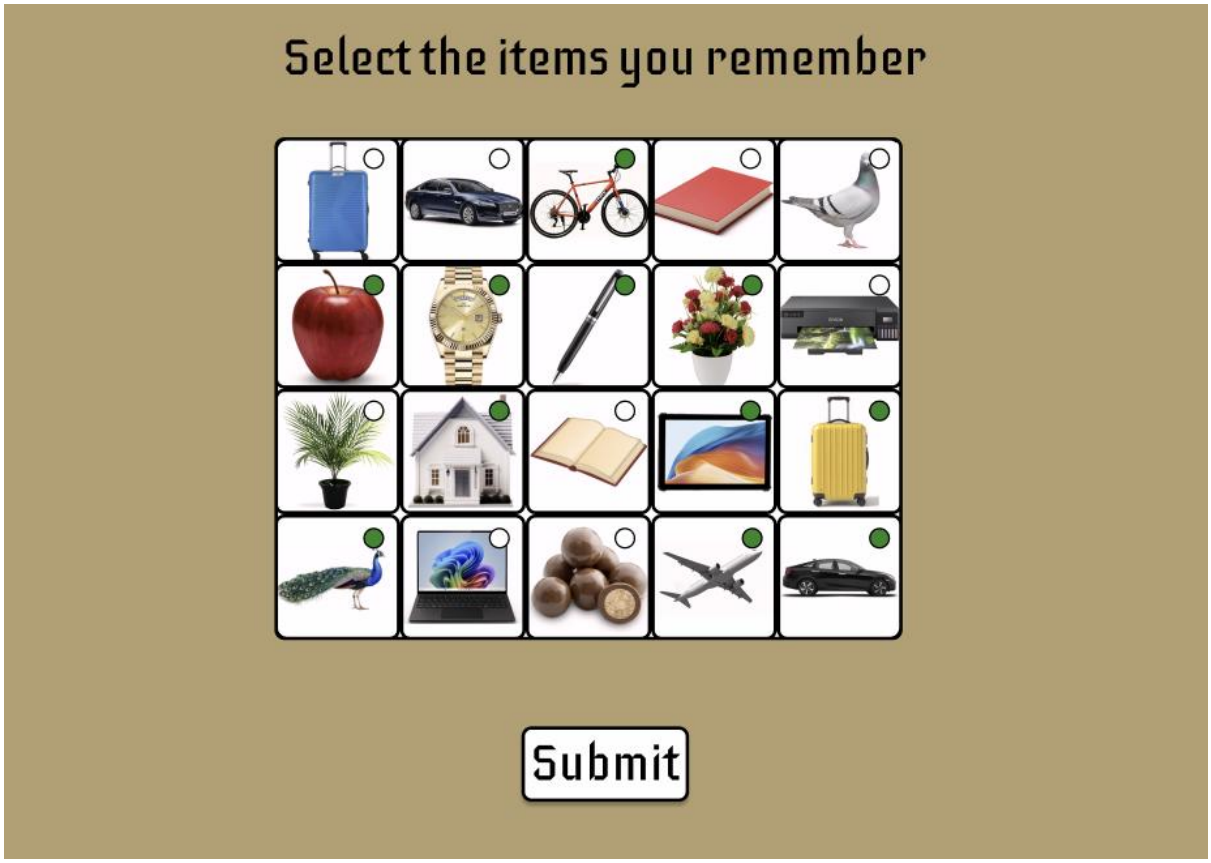
- Clearly visible and placed at the bottom for intuitive completion of the task.

3. How the Recall Phase Works

1. Users recall icons from memory.
2. They select all icons they believe were shown earlier.
3. No penalty is applied for wrong selections.
4. Each correct selection earns 1 mark.
5. Clicking “Submit” moves the user to the results screen.

4. Cognitive and UX Benefits

- Distractors test memory accuracy rather than guessing.
- No negative marking reduces stress and encourages honest recall.
- Clean layout ensures focus remains on decision-making.



FRAME 4: RESULT PAGE

Analysis of the Result and Feedback Screen

This screen provides performance feedback and completes the memory recall cycle.

1. Purpose of the Screen

- Displays the user's recall score clearly.
- Provides motivational feedback.
- Allows users to restart the task.

2. Key UI Elements

Score Display

- Example: "Your Score: 7/10"
- Shows the number of correct selections out of total correct icons.

Feedback Message

- Example: "Good attempt! You recalled most of the items correctly."
- Positive reinforcement improves user experience.

Restart Button

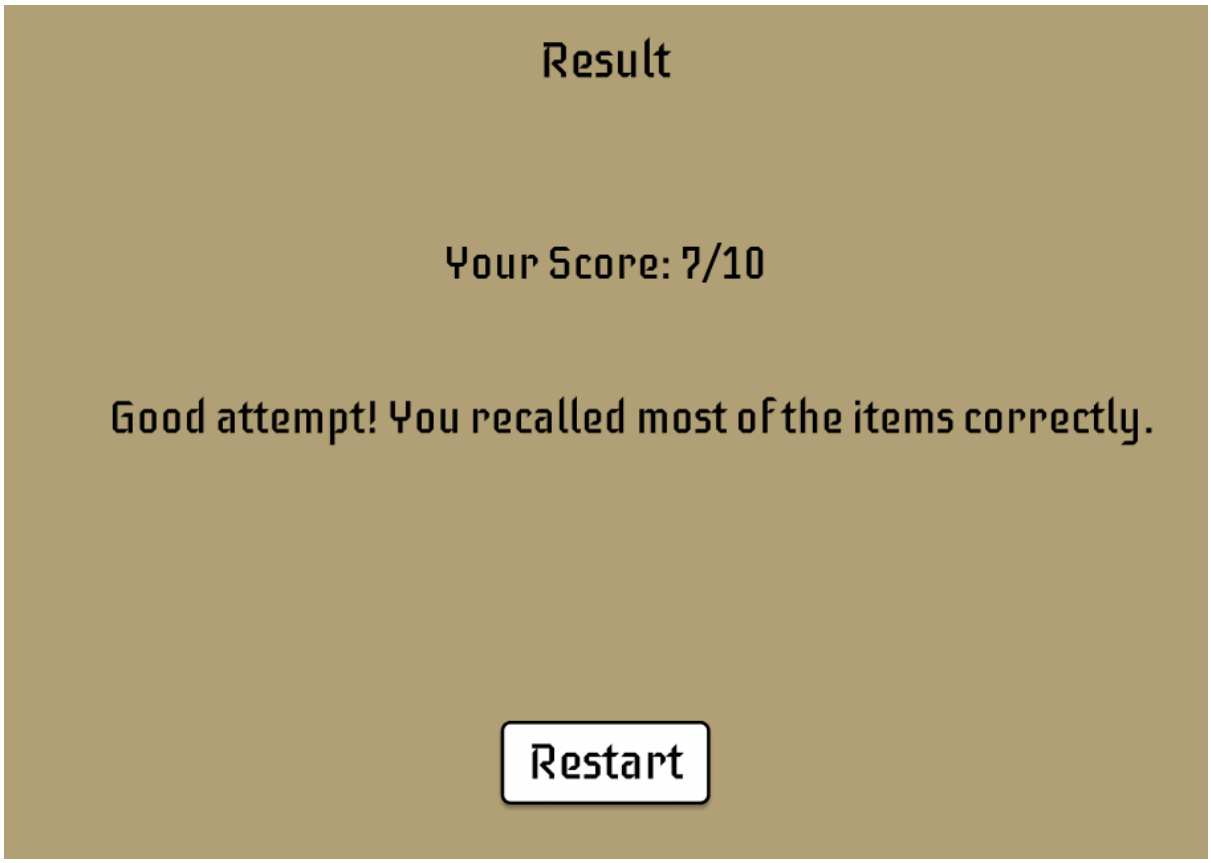
- Allows users to retake the task and improve performance.

3. How This Phase Works

1. The system compares selected icons with the original set.
2. Each correct icon earns 1 point.
3. The final score is displayed instantly.
4. Users can choose to restart and try again.

4. Cognitive and UX Benefits

- Immediate feedback strengthens learning.
- Simple score presentation avoids confusion.
- Encourages repeated attempts, reinforcing memory skills.



PROTOTYPE LINK

https://www.figma.com/proto/6IzxsJiw8LkDiHa04kBJqk/Visual-Memory-Recall-Task?page-id=34%3A35&node-id=34-36&p=f&viewport=25%2C288%2C0.14&t=bPQo10KlfOnKTXRU-1&scaling=min-zoom&content-scaling=fixed&starting-point-node-id=34%3A36